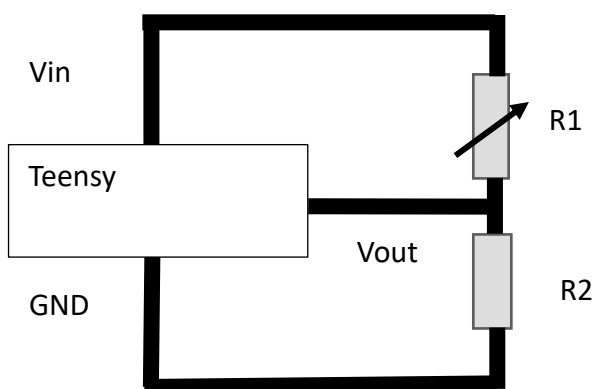
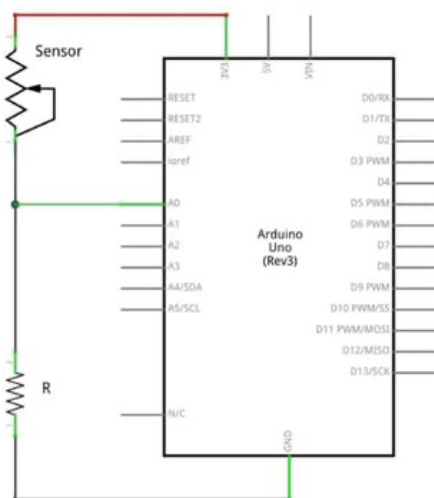


- **V_{in}**: The input voltage source.
- **R1**: A fixed resistor connected in series.
- **R2**: A variable resistor (or potentiometer) which can be adjusted to change the resistance.
- **V_{out}**: The voltage measured across the variable resistor, which can be adjusted by changing the position of the potentiometer.
- **GND**: The ground reference point for the circuit



Basic Components

1. **Resistors (R_1 and R_2):**
 - These are the two resistors used in the voltage divider.
 - R_1 is connected to the input voltage source (V_{in}), while R_2 is connected between R_1 and ground.
2. **Input Voltage (V_{in}):**
 - This is the voltage you want to divide. It's connected to the top of the voltage divider (across R_1).
3. **Output Voltage (V_{out}):**
 - This is the voltage taken from the junction between R_1 and R_2 (the node where they connect). It's the divided voltage you want to obtain.
4. **Ground:**
 - This is the reference point in the circuit, usually at 0 volts. It is connected to the lower end of R_2 , completing the circuit.